KSJ RESEARCH

Research Note Vol.3

Pricing an IPO in a 40x P/E Sector

An indicative price range for a hypothetical power-equipment maker — multiples, discounts and cycle risk from Korean and U.S. peers

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Indicative range KRW 66,200–79,500 (current basis) vs KRW 46,900–57,600 (with peak margins) — a 10–40% discount alone cannot cover the double peak

Three-step peer selection → multiple (P/E main, EV secondary) → value → discount → range → Monte Carlo on both the peak multiple and the peak margin

Key metrics [*1,*2,*3,*4]

Key metric	Value	p.
Applied P/E current · normalised · conservative	**41.4x · 28.5x · 20.1x**	5
3 Korean names aggregate P/E (mean · median)	40.1x (48.9x · 34.7x)	4
U.S. core peers' P/E (Vertiv, Eaton)	42.7x	4
EV/EBIT (secondary, 3 Korean names)	28.6x	4
Target Co. NI to parent FY2025A (assumed)	KRW 30.8bn	2
Fair equity value (current multiple)	KRW 1,273bn	6
Value per share (post-IPO shares)	**KRW 101,810**	6
Range · current basis (discount 22–35%)	**KRW 66,200–79,500**	6
Range · double peak (discount 43–54%)	**KRW 46,900–57,600**	7
Value: normalised earnings × multiple	KRW 47,950	5
Offer size (range low–top)	KRW 207bn–KRW 248bn	6
1-yr value P10 · P50 · P90 (current basis)	70,515 · 87,224 · 107,275	7
1-yr loss probability at 40% (current · double peak)	1% · 58%	7

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares *4 KSJ Research
Note: KSJ Research. Target Co. is hypothetical; definitions and sources in footnotes and A2. Judgements are the author's own

41.4x

Applied P/E (Korea and U.S. core average, as of)

41.4x is an earnings-made multiple, but a peak one — normalised 28.5x, conservative 20.1x

- Normalised 28.5x: log midpoint of early-2023 20.1x and today (half the re-rating reverses)
- Vol.1: 61% of the same names' gain was earnings — the multiple itself signals a peak
- LS ELECTRIC at 86.1x exceeds 2x the median — kept under aggregate weighting (mean would be 48.9x)
- GE Vernova held pending its 10-K — including it gives U.S. core 45.9x (now 42.7x)

KRW 101,810

Value per share (current multiple, post-IPO shares)

Value KRW 101,810 → indicative range KRW 66,200–79,500 (median Korean IPO discount, sub-sample (KOSPI or offer ≥ KRW 100bn), n = 19)

- NI KRW 30.8bn × 41.4x = equity KRW 1,273bn ÷ 12.5mn shares post-IPO
- Offer 3.12mn shares (80% primary, 20% secondary) → KRW 207bn–KRW 248bn
- 10–40% discount scenarios: KRW 61,086–KRW 91,629
- EV/EBIT cross-check: KRW 125,251 per share (+23% vs P/E)

43–54%

Required discount with a double peak (1-yr value P50 · P25)

The price of a double peak: 43%+ — outside the 10–40% grid; a discount alone does not cover it

- Current basis (margin fixed at 10.3%): 14%–23% → KRW 66,200–79,500
- Letting margin range FY2022A 3.1%–FY2025A 10.3%: 43%–54% → KRW 46,900–57,600
- Even at 40% the 1-yr loss probability is 58% (current basis 1%)
- 97% of H2 2025 IPOs priced at the top (FSS) — the discount is set when the range is designed

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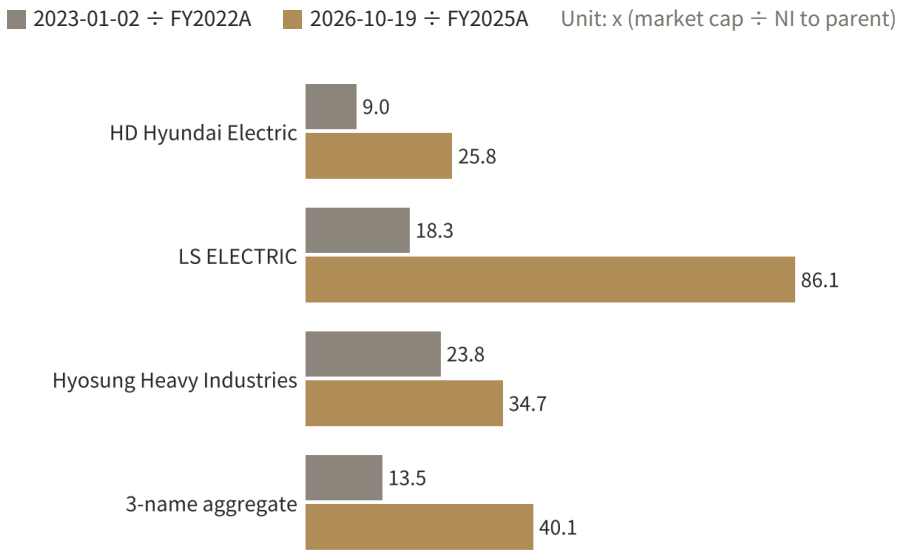
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Starting point: P/E 13.5x → 40.1x, 61% of the gain was earnings — earnings that were developers' costs

Vol.1 earnings-led re-rating + Vol.2 equipment pricing power → what range for a hypothetical unlisted company at this multiple?

3-name aggregate P/E 13.5x (early 2023) → 40.1x (as of) — 61% of the gain was earnings *1,*2,*3



Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 KSJ Research Vol.1 (Oct 2026)
Note: Vol.1 log split: Δln(cap) = Δln(NI) + Δln(P/E). Doosan Enerbility excluded (FY2022A net loss)

Vol.2: peer margin 5.2% → 14.4% — if all from price, equipment prices +10.9% (upper bound), the developer's cost *1,*2

Company	OPM FY2022A	OPM FY2025A	Implied price rise
HD Hyundai Electric	6.3%	24.4%	23.9%
LS ELECTRIC	5.6%	8.6%	3.3%
Hyosung Heavy Industries	4.1%	12.5%	9.6%
3-name total	5.2%	14.4%	**10.9%**

Source: *1 FSS DART annual reports (consolidated) *2 KSJ Research Vol.2 (Oct 2026)
Note: Implied rise = (1 − OPM start) ÷ (1 − OPM end) − 1: an upper bound with unit cost unchanged (not causal). That profit is today's P/E denominator

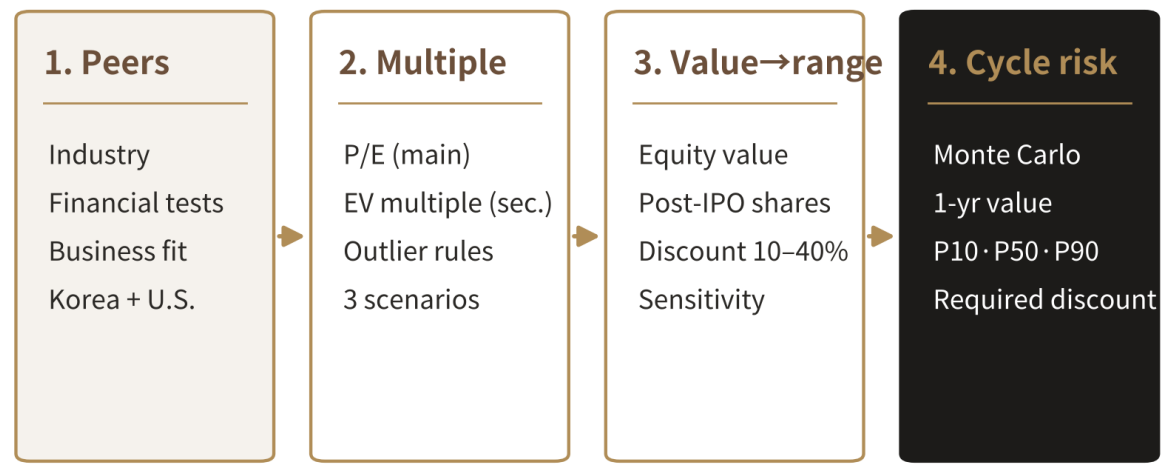
Target Co.: a hypothetical unlisted power-equipment maker

Question: can an earnings-made 40x P/E be applied to an unlisted company's IPO price?

- Hypothetical (not a real company): a Korean mid-size maker of switchgear, UPS and data-centre power distribution
- FY2025A revenue KRW 300bn (design) — growth, margins and net debt from Korean peers' DART data
- 10mn shares pre-IPO · offer 25% of post-IPO shares (80% primary) → 12.5mn shares
- As of 2026-10-20 — caps at the prior trading day, earnings FY2025A (latest fiscal year)

Note: Assumptions and their status on p2 and A2

Korean IPO practice: value → discount → indicative range *1



Source: *1 KSJ Research
Note: A generic structure, not a copy of any filing; final pricing (book-building) is out of scope

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Target Co. inputs are Korean peers' DART aggregates — growth 18.6%, operating margin 14.4%, net margin 10.3%

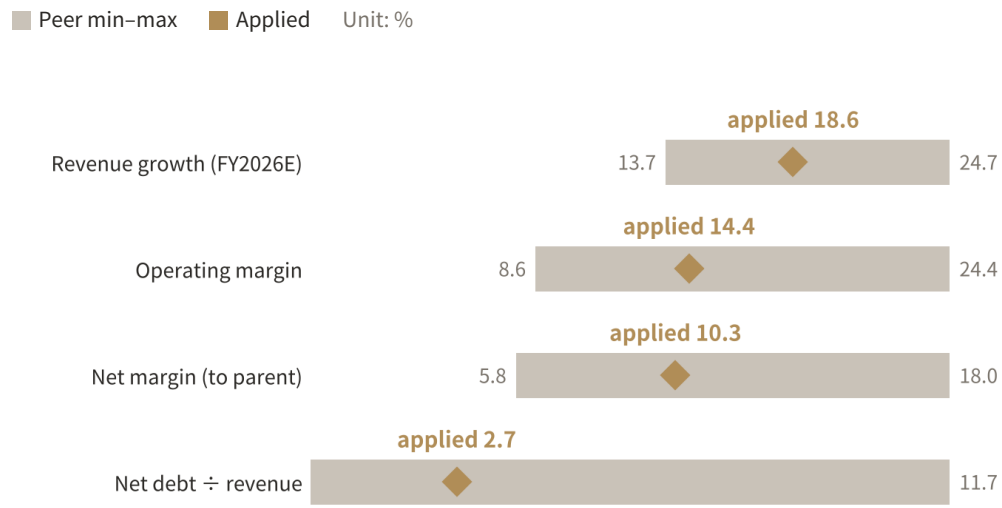
Revenue KRW 300bn and 10mn shares pre-IPO are design values (price unit); the rest are peer aggregates within the peer range

4 of 9 Target Co. inputs come from Korean peers' DART data — ranges are the min–max of 3 names *1,*2,*3

Item	Applied	Peer range	Status
Revenue FY2025A	3,000	—	Design
Revenue growth (FY2026E)	18.6%	13.7%–24.7%	Data
Operating margin	14.4%	8.6%–24.4%	Data
Net margin (to parent)	10.3%	5.8%–18.0%	Data
Net debt ÷ revenue	2.7%	-18.7%–11.7%	Data
D&A ÷ revenue	—	—	Input needed
Pre-IPO shares	10,000,000	—	Design
Offer size (% of post-IPO shares)	25.0%	—	Verify
Primary share (rest secondary)	80.0%	—	Design

Source: *1 FSS DART annual reports (consolidated) *2 Framework Act on SMEs Enforcement Decree Annex 1; KRX listing rules (free-float requirement) *3 KSJ Research
Note: Applied = 3-name combined (growth = combined revenue CAGR FY2022A→FY2025A). Status: data = DART, design = hypothetical setting, verify = public rule pending check

Applied values (◆) sit inside the peer range — margins from LS ELECTRIC 8.6% to HD Hyundai Electric 24.4% *1



Source: *1 FSS DART annual reports (consolidated)
Note: Range = Korean peers' FY2025A min–max. HD Hyundai Electric: debt = other financial liabilities (current + non-current), a proxy pending the notes

FY2025A NI to parent KRW 30.8bn — the earnings the price is built on *1,*2

Unit: KRW 100mn → bn, KRW

Item	FY2025A	FY2026E (growth applied)
Revenue	KRW 300bn	KRW 356bn
Operating income	KRW 43.3bn	KRW 51.4bn
NI to parent	**KRW 30.8bn**	KRW 36.5bn
EBIT	KRW 43.3bn	—
Net debt	KRW 8.0bn	—
EPS (pre-IPO shares)	KRW 3,075	KRW 3,649

Source: *1 FSS DART annual reports (consolidated) *2 KSJ Research
Note: FY2026E = FY2025A × (1 + 18.6%), margins fixed. The price applies peers' P/E on the same basis (FY2025A)

How inputs move the result: scale is irrelevant; margins and multiples matter

- Revenue scale and share count only set the price unit: value = earnings × multiple; the discount conclusion does not depend on scale
- 1pp of net margin = KRW 3.0bn NI = KRW 124bn of value at the current multiple
- Margins are cyclical too: 3-name net margin FY2022A 3.1% → FY2025A 10.3%, operating 5.2% → 14.4%
- At FY2022A margins, NI on the same revenue would be 31% of today's — multiple and earnings may both be at a peak

Note: The margin cycle is not in the Monte Carlo; see limitations (p8) and Q&A (A1)

Peers: 3 Korean + 2 U.S. core equipment makers; 2 generators as reference — Doosan Enerbility at 503x P/E and Schneider Electric (not an SEC filer) excluded; GE Vernova held pending the 10-K

Prospectus-style three steps: industry (power infrastructure) → financial tests (profit, multiple, filer) → business fit (distribution, UPS, data-centre power)

10 candidates → 5 applied · 1 held — industry → financial tests → business fit *1,*2,*3

Company	1. Industry	2. Financial tests	3. Business fit	Result
HD Hyundai Electric	Power equipment	Profitable · P/E 25.8x	Transformers, HV equipment (medium)	Applied
LS ELECTRIC	Power equipment	Profitable · P/E 86.1x	Switchgear, breakers (high)	Applied
Hyosung Heavy Industries	Power equipment	Profitable · P/E 34.7x	Transformers, breakers (heavy industries) (medium)	Applied
Doosan Enerbility	Generation equip.	P/E 503x > 100x	Nuclear and gas-turbine generation equipment (low)	Excluded
Vertiv	Power equipment	Profitable · P/E 59.5x	Data-centre UPS, power distribution, thermal (high)	Applied
Eaton	Power equipment	Profitable · P/E 37.2x	Power distribution, UPS, power management (electrical) (high)	Applied
GE Vernova	Power equipment	Held: NI/OI 3.5x · EV/EBIT 168x	Gas turbines, wind, grid equipment (Electrification) (high)	Held: pending 10-K
Schneider Electric	Power equipment	Not an SEC filer	Distribution, power management, automation (similar business)	Excluded
Constellation Energy	Power generation	Profitable · P/E 40.2x	Nuclear-led generator (generation)	Reference
Vistra	Power generation	Profitable · P/E 53.2x	Generator and retailer (generation)	Reference

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Financial tests: filed financials, FY2025A profit, P/E ≤ 100x, listed entity = filer (SEC). Business notes summarise annual-report segments (author)

Equipment makers and generators are different businesses — net debt ÷ revenue 28% vs 58% *1

Company	Group	Rev. CAGR 23→25	OPM	Net margin	Net debt ÷ rev.
Vertiv	Core · equipment	22.1%	17.9%	13.0%	12%
Eaton	Core · equipment	8.8%	—	14.9%	34%
GE Vernova	Core · held	7.0%	3.6%	12.8%	-23%
Constellation Energy	Ref. · generation	4.3%	13.6%	10.2%	24%
Vistra	Ref. · generation	12.9%	10.8%	5.4%	103%

Source: *1 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: SEC 10-K FY2025A (USD); revenue CAGR FY2023A→FY2025A. Equipment = orders and production, margins from pricing power / generation = power sales and PPAs with heavy assets and debt, so P/E and EV behave differently → reference only. Eaton reports no operating-income tag (—); group net-debt totals exclude held names

Market cap = close × SEC shares — 0 outside ±2.0% of yfinance shares *1,*2,*3

Company	Close USD	SEC shares (mn)	yfinance (mn)	Diff.	Market cap	In KRW
Vertiv	206.08	385.0	385.0	+0.0%	USD 79,338mn	KRW 105,687bn
Eaton	391.50	388.4	388.4	+0.0%	USD 152,059mn	KRW 202,557bn
GE Vernova	908.44	266.3	266.3	+0.0%	USD 241,948mn	KRW 322,299bn
Constellation Energy	263.19	354.3	354.3	+0.0%	USD 93,250mn	KRW 124,219bn
Vistra	149.70	335.6	335.6	+0.0%	USD 50,245mn	KRW 66,931bn

Source: *1 yfinance (Yahoo Finance, secondary; cross-checked with SEC shares) *2 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares *3 Bank of Korea ECOS USD/KRW
Note: yfinance is secondary — SEC cover shares are used; the difference is checked only. KRW = USD mn × 1,332.1 ÷ 100 (ECOS 2026-10-20)

Exclusions and checks: distorted multiple, disclosure scope, filer identity

- Doosan Enerbility: P/E 503x (denominator distortion from FY2025A NI of KRW 84.8bn) and low fit (generation equipment). Also excluded from Vol.1's split (FY2022A loss)
- Schneider Electric: Not an SEC filer (Euronext Paris) — manual-entry template only; excluded this volume
- GE Vernova: excluded from P/E and EV multiples pending the 10-K (kept in tables) — FY2025A NI USD 4,884mn = 3.5x operating income USD 1,388mn (possible one-offs); No long-term-debt tag → net debt USD -8,785mn (net cash may be overstated); EV/EBIT 168x
- Vistra: automatic entity check WARN → OK after checking the 10-K cover registrant (2026-10-22)

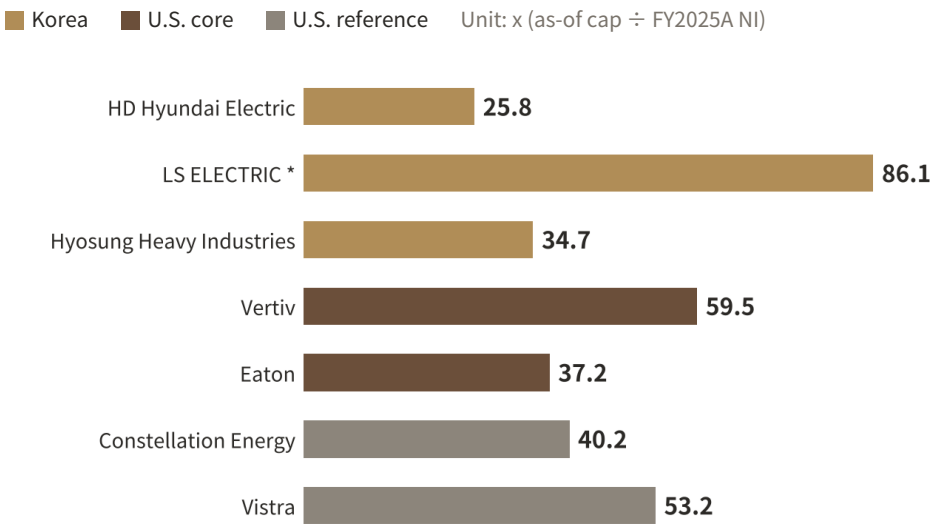
Note: Outlier rules in the p4 table

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Applied P/E 41.4x: Korea aggregate 40.1x (mean 48.9x, median 34.7x) + U.S. core 42.7x

P/E (main) · EV/EBIT (secondary) · outlier rules R1–R5 — Korea, U.S. and combined computed separately

Current P/E: Korea 25.8x–86.1x · U.S. core 37.2x–59.5x — LS ELECTRIC 86.1x exceeds 2x the median (*, kept) *1,*2,*3



Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Korea = KRX 2026-10-19 cap ÷ DART FY2025A NI to parent. Excluded names not shown (p3)

Applied P/E 41.4x — aggregate within groups, average across groups *1,*2,*3

Group	n	Aggregate	Mean	Median	Applied
3 Korean names	3	40.1x	48.9x	34.7x	40.1x
U.S. core (equipment)	2	42.7x	48.4x	48.4x	42.7x
U.S. reference (generation)	2	44.0x	46.7x	46.7x	compare only
Applied (Korea and U.S. core average)					**41.4x**

Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Aggregate = Σcap ÷ ΣNI. With 3–5 peers a simple mean is driven by one name, so the aggregate leads, mean and median shown. Korea and U.S. are averaged as groups (very different sizes)

Outlier rules: drop losses and extreme multiples; flag, but keep, high ones *1

Rule	Test	Treatment	Hit
R1	NI to parent ≤ 0	Drop P/E	—
R2	P/E > 100x	Drop (denominator distortion)	Doosan Enerbility
R3	Listed entity ≠ filer (SEC)	Drop	—
R4	> 2x group median	Flag, keep (aggregate)	LS ELECTRIC
R5	EBITDA ≤ 0, EV multiple > 60x, or no net debt	Drop EV multiple	—
R6	Earnings quality / debt-tag doubts (NI > 1.5x OI etc.)	Drop P/E and EV pending the 10-K (kept in tables)	GE Vernova

Source: *1 KSJ Research
Note: Rules set before looking at the multiples (author's judgment); the same rules apply to the early-2023 point

EV/EBIT 36.3x (secondary) — applied to Target, value is +23% vs P/E *1,*2,*3

Company	EV	EBIT	Multiple	Treatment
HD Hyundai Electric (proxy)	KRW 18,160bn	KRW 995bn	18.2x	Kept
LS ELECTRIC	KRW 25,270bn	KRW 426bn	59.3x	Kept
Hyosung Heavy Industries	KRW 18,638bn	KRW 747bn	25.0x	Kept
Vertiv	USD 80,523mn	USD 1,830mn	44.0x	Kept
Eaton	USD 161,332mn	—	—	No EBIT
GE Vernova	USD 233,163mn	USD 1,388mn	168.0x	R6 held

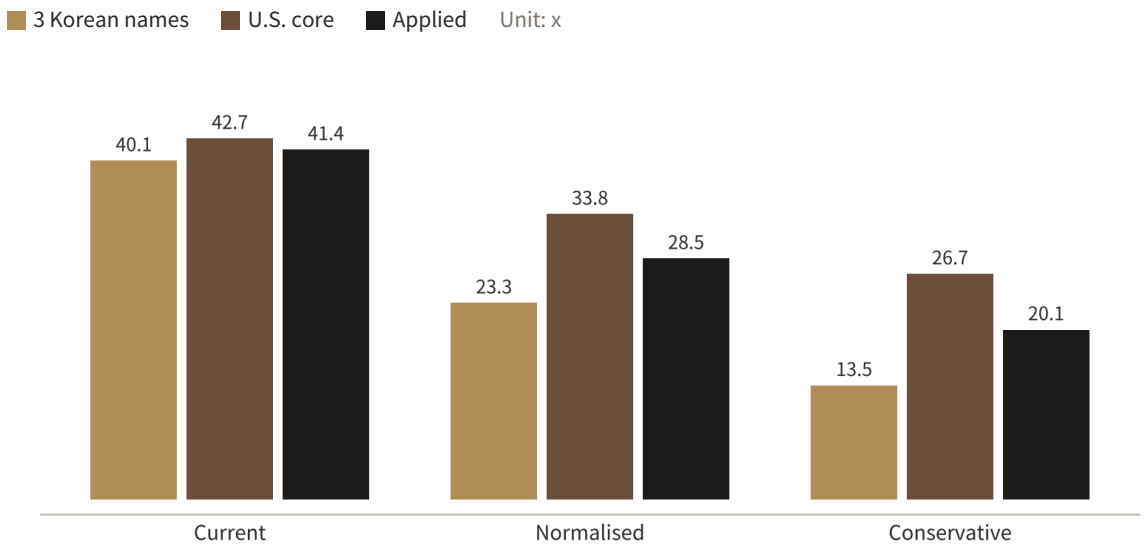
Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: D&A missing (HD Hyundai Electric, Hyosung Heavy Industries) → EBIT stands in for EBITDA (EV/EBIT ≥ EV/EBITDA), applied to Target's EBIT for consistency; switches to EV/EBITDA automatically once all applied peers have D&A. EV = cap + net debt (ex leases). Aggregate n = 3

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Scenarios: current 41.4x · normalised 28.5x · conservative 20.1x — normalise earnings too and value is 47% of the current basis

Conservative = observed early-2023 P/E (same peers, same rules); normalised = log midpoint — outlier rules applied at both points

Current 41.4x → normalised 28.5x → conservative 20.1x: equity value falls to 69% and 49% *1,*2,*3



Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Scenarios are a range, not probabilities; the Monte Carlo (p7) uses a triangular distribution from conservative to current

Normalise earnings too and value is KRW 47,950 per share — 47% of the current-basis value *1,*2,*3

Scenario	Definition	Korea	U.S. core	Applied	Value/share
Current	As-of (2026-10-19) cap ÷ FY2025A	40.1x	42.7x	**41.4x**	KRW 101,810
Normalised	√(conservative × current): 50% log reversal	23.3x	33.8x	**28.5x**	KRW 70,129
Conservative	First 2023 trading day (2023-01-02) cap ÷ FY2022A	13.5x	26.7x	**20.1x**	KRW 49,432
Normalised earnings × multiple	Net margin 7.0% (4-yr cycle average) × normalised multiple	—	—	28.5x	**KRW 47,950**

Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Applied = simple average of Korea and U.S. core group multiples. Last row = earnings also back to the cycle average (both peaks unwound) — same revenue and shares

By company: Korea early-2023 P/E 9.0x–23.8x → today *1,*2,*3

Company	Cap 2023-01-02	NI FY2022A	P/E early-23	P/E now	Change
HD Hyundai Electric	KRW 1,458bn	KRW 162bn	9.0x	25.8x	2.9x
LS ELECTRIC	KRW 1,653bn	KRW 90.3bn	18.3x	86.1x	4.7x
Hyosung Heavy Industries	KRW 693bn	KRW 29.1bn	23.8x	34.7x	1.5x
Doosan Enerbility	KRW 9,734bn	KRW -772bn	R1	503.3x	—
Vertiv	USD 5,018mn	USD 77mn	65.5x	59.5x	0.9x
Eaton	USD 62,737mn	USD 2,462mn	25.5x	37.2x	1.5x
GE Vernova	—	USD -2,736mn	—	49.5x (held)	—

Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: R1 = FY2022A net loss → also excluded at the conservative point. U.S. = first-2023 close × latest prior SEC cover shares ÷ FY2022A NI; held (R6) names excluded at both points

Why a log midpoint: the same scale as Vol.1's split

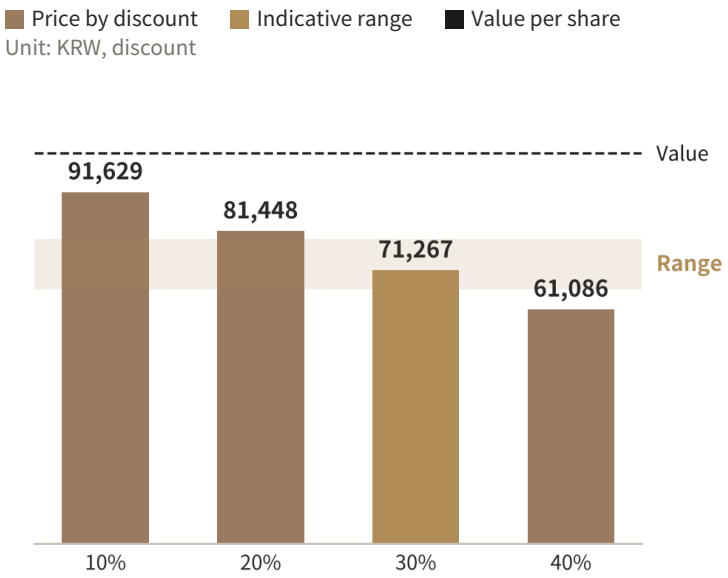
- Vol.1 split the gain in logs: earnings 61%, multiple 39%
- Normalised = 50% of the re-rating reverses on the same log scale → 28.5x
- An arithmetic midpoint would be 30.7x — tilted toward the high multiple
- Equity value: current KRW 1,273bn · normalised KRW 877bn · conservative KRW 618bn
- Not only the multiple: if net margin also returns from 10.3% to the 4-yr average 7.0%, equity value is KRW 599bn (normalised × normalised)

Note: The 50% reversal is the author's judgment; the Monte Carlo spans conservative to current

Value KRW 101,810 per share → indicative range KRW 66,200–79,500 (22–35% discount) — offer KRW 207bn–KRW 248bn

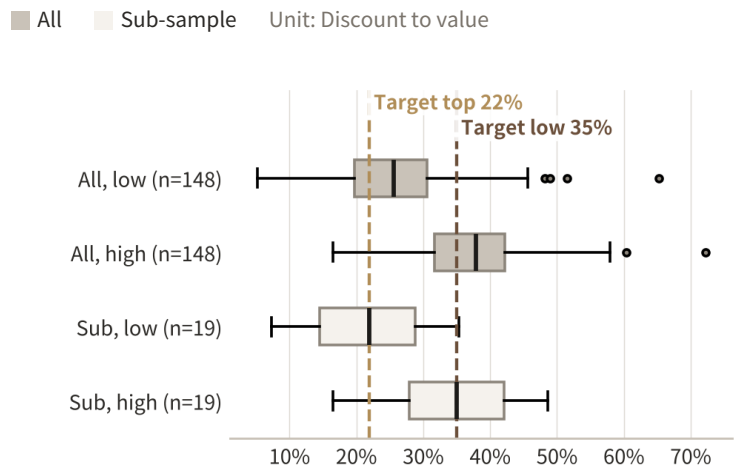
Equity value → post-IPO shares (primary, secondary) → value per share → discount scenarios (10–40%) → range · sensitivity (multiple × discount)

10–40% discount: KRW 61,086–KRW 91,629 — the range sits at 22–35% *1



Source: *1 KSJ Research
Note: Discount = 1 − price ÷ value per share (current multiple)

Korean IPO discounts (sub-sample (KOSPI or offer ≥ KRW 100bn), n = 19): median 22% (low) · 35% (high) — Target's range at P53 *1



Source: *1 DART equity prospectuses, new listings filed 2024-12-03 to 2026-10-20
Note: DART equity prospectuses for new listings, filed 2024-12-03 to 2026-10-20; first filing with a price range per company. Target (offer ≈ KRW 207bn) uses the sub-sample (n ≥ 10). IQR low 15–29%, high 28–42%. Outliers (high > 50% or low < 10%, to check) 2 — excluding them, medians 23.6% / 38.9%.

Sensitivity: 5x of P/E = KRW 12,302 per share (before discount) — cells inside the range in gold *1

Unit: KRW (rows = applied P/E, columns = discount)

Applied P/E	10%	20%	30%	40%
Current 41.4x	**91,629**	**81,448**	**71,267**	**61,086**
Normalised 28.5x	**63,116**	**56,103**	**49,090**	**42,077**
Conservative 20.1x	**44,488**	**39,545**	**34,602**	**29,659**
15x	33,214	29,524	25,833	22,143
20x	44,286	39,365	34,445	29,524
25x	55,357	49,207	43,056	36,905
30x	**66,429**	59,048	51,667	44,286
35x	**77,500**	**68,889**	60,278	51,667
40x	88,572	**78,731**	**68,889**	59,048
45x	99,643	88,572	**77,500**	**66,429**

Discount

Source: *1 KSJ Research
Note: Top three rows = scenarios (p5), below = 5x grid. Cells above the range in warm grey

Value KRW 101,810 = KRW 30.8bn × 41.4x ÷ 12.5mn shares → range KRW 66,200–79,500 *1

	Step	Value
①	Earnings: Target Co. FY2025A NI to parent	KRW 30.8bn
②	× applied P/E (current)	41.4x
③	= Fair equity value	KRW 1,273bn
④	÷ post-IPO shares (10mn shares + 2.5mn shares new)	12.5mn shares
⑤	= Value per share	**KRW 101,810**
⑥	Discount (median Korean IPO discount, sub-sample (KOSPI or offer ≥ KRW 100bn), n = 19)	22–35%
⑦	= Indicative range (rounded to KRW 100)	**KRW 66,200–79,500**
⑧	Ref.: double-peak range (43–54% discount, p7)	KRW 46,900–57,600

Source: *1 KSJ Research
Note: Value per share uses post-IPO shares (new-share proceeds not added — conservative). Range discounts = medians of Korean IPO prospectus discounts (low and high)

Offer 3.12mn shares (80% primary) · KRW 207bn–KRW 248bn — existing holders' 75% is the overhang *1,*2,*3

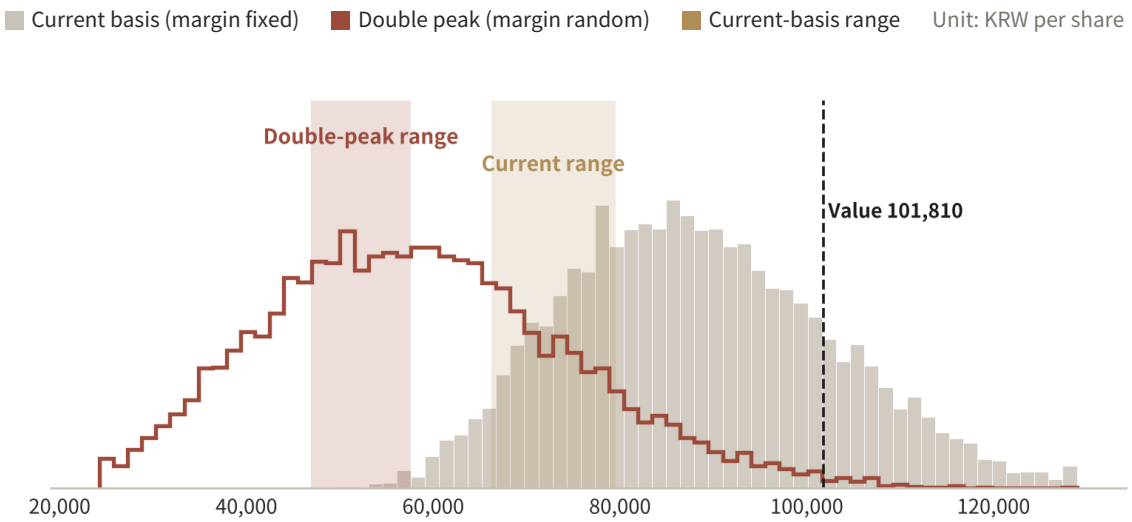
Item	Shares	Amount (range low–top)
Pre-IPO shares	10,000,000	—
Offer shares (25% of post-IPO)	3,125,000	KRW 207bn–KRW 248bn
Primary (to company)	2,500,000	KRW 166bn–KRW 199bn
Secondary (to sellers)	625,000	KRW 41.4bn–KRW 49.7bn
Post-IPO shares	12,500,000	Cap KRW 828bn–KRW 994bn
Existing holders' stake (overhang)	9,375,000	75.0%
Dilution from new shares		20.0%

Source: *1 Framework Act on SMEs Enforcement Decree Annex 1; KRX listing rules (free-float requirement) *2 Money Today (4 Mar 2026, citing an FSS release): 2025 IPO lock-up commitments and pricing within the range *3 KSJ Research
Note: Offer size at the listing rules' free-float level (to verify); primary share is hypothetical. Major-holder lock-up not set → shares only. Ref.: institutional lock-up commitments 41% in 2025 (prior year 18%); KOSPI 54% — FSS

Current basis 14%–23% vs double peak 43%–54% — adding peak margins raises the needed discount by 29pp

10,000 runs × 3 (current basis · double peak · co-move sensitivity) · seed 42 · 1-yr growth × multiple (conservative–current) × margin (FY2022A–FY2025A)

With margins also at a peak, 1-yr P50 falls 87,224 → 57,641 (-34%) — the KRW 101,810 value sits at P99 of the double-peak distribution *1,*2,*3,*4



Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares *4 KSJ Research
Note: 1-yr value = FY2025A NI × (1 + growth) × 1-yr P/E × (1-yr margin ÷ current margin) ÷ post-IPO shares. 10,000 runs each, same seed

Even at a 40% discount the double-peak 1-yr loss probability is 58% — median break-even discount 14% → 43% *1

Metric	Current basis	Double peak	Co-move ρ=0.5
1-yr value P10	70,515	38,436	34,197
1-yr value P25	77,953	46,934	44,232
1-yr value P50	87,224	57,641	58,012
1-yr value P90	107,275	79,692	90,391
Needed discount (P50)	14%	43%	43%
Needed discount (P25)	23%	54%	57%
Needed discount (P10)	31%	62%	66%
Loss prob. at 10%	61%	97%	91%
Loss prob. at 20%	34%	91%	83%
Loss prob. at 30%	11%	79%	71%
Loss prob. at 40%	1%	58%	55%
Range	KRW 66,200–79,500	KRW 46,900–57,600	

Source: *1 KSJ Research
Note: Needed discount = 1 − quantile ÷ current-basis value; loss probability = P(1-yr value < price). Red = above the 40% grid top. Co-move = pairwise correlation 0.5 among growth, P/E and margin (sensitivity)

Convergence: Double peak, Co-move ρ=0.5 moved more than 1% from 5,000 to 10,000 runs — batch s.e. ≤ 0.73%, ≤ 0.5% from a 10x rerun *1

Run	P50 at 5k	P50 at 10k	Change	Verdict	Batch s.e.	P50 at 100,000
Current basis	87,103	87,224	0.14%	Yes	0.23%	86,969 (-0.3%)
Double peak	56,905	57,641	1.29%	Above tol.	0.46%	57,374 (-0.5%)
Co-move ρ=0.5	57,224	58,012	1.38%	Above tol.	0.73%	57,720 (-0.5%)

Source: *1 KSJ Research
Note: Verdict = relative P50 change between the first 5,000 and 10,000 draws ≤ 1%. Reported values stay at 10,000 runs even when above tolerance; the error size is shown via a 10-batch standard error and a 100,000-run rerun

Inputs: growth and multiple, plus net margin, so peak earnings and peak multiples move together *1,*2,*3

Input	Distribution / value	Basis
Earnings growth (1 yr)	ln(1+g) ~ N(0.175, 0.063)	9 Korean peer revenue-growth observations
1-yr-ahead P/E	Tri(20.1, 28.5, 41.4)	Min conservative · mode normalised · max current (neutral)
1-yr net margin (double peak)	Tri(3.1%, 7.0%, 10.3%)	FY2022A · 4-yr cycle average (mode; mean-reversion judgment) · FY2025A — 3 Korean names
Correlation (growth, P/E, margin)	Base 0.0/0.0/0.0 · co-move 0.5	Gaussian copula; no data to estimate → independent base (neutral); joint downturn as sensitivity
Runs · seed	10,000 · 42	Reproducible
Range basis	P25 · P50	Low 25% · top 50% loss probability (judgment)

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 KSJ Research
Note: Current basis = margin fixed (same as before). Double peak = margin also random. Zero correlation assumes slowdown, de-rating and margin compression arrive separately

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Implication: a double peak (peak earnings × peak multiple) is not covered by a 10–40% discount alone — earnings durability and deal structure must do the rest

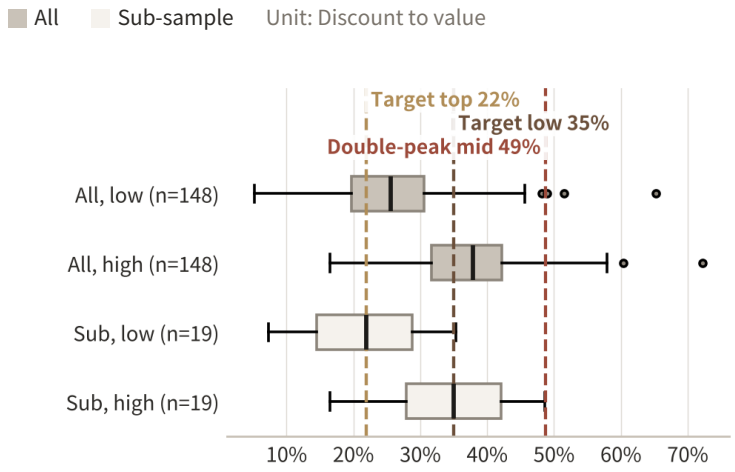
Conclusion (risks a discount covers vs does not) · ECM checklist · limitations · next: Vol.4 (data-centre power supply)

Conclusion: the discount covers the peak multiple; the peak margin needs structure and disclosure *1,*2

- Current-basis range KRW 66,200–79,500 (22–35% discount) vs double-peak KRW 46,900–57,600 (43–54%)
- The double-peak median break-even discount of 43% exceeds the 40% grid top — even at 40% the 1-yr loss probability is 58%: a discount alone does not cover it
- In 2025 100% of IPOs priced within the range and 97% of H2 at the top — the effective discount is the top's (22%)
- Risk the discount cannot carry goes into structure: evidence of margin durability (backlog, long-term contracts), less secondary sale, lock-ups

Source: *1 Money Today (4 Mar 2026, citing an FSS release): 2025 IPO lock-up commitments and pricing within the range *2 KSJ Research
Note: From the Monte Carlo (p7) · pricing positions from an FSS release

Korean IPO discounts (sub-sample (KOSPI or offer ≥ KRW 100bn), n = 19): median 22% (low) · 35% (high) — Target's range at P53 *1



Source: *1 DART equity prospectuses, new listings filed 2024-12-03 to 2026-10-20
Note: DART equity prospectuses for new listings, filed 2024-12-03 to 2026-10-20; first filing with a price range per company. Target (offer ≈ KRW 207bn) uses the sub-sample (n ≥ 10). IQR low 15–29%, high 28–42%. Outliers (high > 50% or low < 10%, to check) 2 — excluding them, medians 23.6% / 38.9%.

Next in Vol.4: powering data centres — generation assets and PPAs

After equipment, the power itself: who sells electricity to data centres, and under what contracts?

- Valuing generators (Constellation, Vistra — this volume's reference peers) and PPA structures
- How PPA price and tenor feed Vol.2's power cost and debt capacity
- Vol.5 finale: an AI energy asset investment map — demand, equipment, PF, IPO and generation on one page

Note: Timing and method of Vol.4–5 may change

ECM checklist: which earnings, which multiple, and how much discount *1,*2

Item	This note	Check in a real deal	Pages
Peer set	3 Korean + 2 U.S. core, 2 reference · GE Vernova held	Check held names' 10-K notes (one-offs, debt); add Schneider from its annual report	3, 4
Applied multiple	P/E 41.4x (Korea 40.1x · U.S. core 42.7x)	Whether the R4-flagged name stays moves the mean a lot	4
Earnings basis	FY2025A net margin 10.3% = highest in 4 years	Check LTM results and backlog margins for a peak	2, 5
Discount	**Current 22–35% · double peak 43–54%**	Compare with Korean IPO prospectus discounts	6, 7
Final pricing	2025: 100% within range, 97% of H2 at the top	Range top ≈ final price → design around the top's discount	8
Secondary · overhang	Secondary 20% · existing 75%	Major-holder lock-up; institutional commitments (41% in 2025)	6

Source: *1 Money Today (4 Mar 2026, citing an FSS release): 2025 IPO lock-up commitments and pricing within the range *2 KSJ Research
Note: Checklist for the hypothetical company; figures from the body; final-pricing data from an FSS release (Money Today, 4 Mar 2026)

Limitations of this note

- Target Co. is hypothetical: size, shares and structure are design values; growth and margins are Korean peer aggregates
- GE Vernova held until its 10-K notes are checked; including it changes the U.S. core multiple
- Some Korean D&A missing → EV/EBIT instead of EV/EBITDA (secondary)
- Earnings are FY2025A (~10 months old); LTM would change multiple and earnings
- Margin distribution (mode = 4-yr average), zero correlation, range quantiles and the 50% reversal are judgments — shown with a co-move sensitivity

Note: Updated with the same code as data arrive

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Q&A: questions on the multiple, the discount and the structure

Questions likely in deal work (valuation, prospectus, investor meetings), with page references

Eight questions an ECM desk would ask [*1,*2,*3,*4,*5]

Question	Answer (from the body)	p.
Shouldn't LS ELECTRIC at 86.1x be dropped?	Flagged (R4, > 2x median), but the aggregate is earnings-weighted: 40.1x with vs 29.5x without. A simple mean would be 48.9x	4
Why is Doosan Enerbility excluded?	P/E 503x (NI KRW 84.8bn) is a denominator distortion (R2), and its nuclear/gas-turbine mix fits poorly	3
Why P/E first and EV/EBITDA second?	IPO prices are set on equity value (NI to parent), and net-debt/D&A data are incomplete. The EV/EBIT cross-check is +23% vs P/E	4
Why is GE Vernova left out?	FY2025A NI USD 4,884mn = 3.5x operating income USD 1,388mn (possible one-offs); No long-term-debt tag → net debt USD -8,785mn (net cash may be overstated); EV/EBIT 168x — held until the 10-K notes are checked (kept in tables). Including it: U.S. core 45.9x vs 42.7x without	3, 4
Why are Constellation and Vistra only references?	They sell power: profits depend on prices and PPAs, and heavy assets and debt make P/E and EV behave unlike equipment makers (p3 net debt ÷ revenue)	3
Isn't using peak peer margins aggressive?	So the margin was re-run across FY2022A 3.1%–FY2025A 10.3% (double peak): needed discount 14% → 43% (P50). Normalised earnings × multiple gives KRW 47,950	5, 7
Then why not a 43% discount?	With a double peak, even a 40% discount leaves a 58% 1-yr loss probability (current basis 1%). Issuers rarely accept discounts beyond the grid and 2025 pricing clustered at the top — the rest is carried by margin disclosure, smaller secondary and lock-ups	7, 8
How should the secondary sale and overhang be read?	20% of the offer is secondary (5.0% of post-IPO). Existing holders' 75% is overhang once lock-ups end — institutional lock-up commitments 41% in 2025 (prior year 18%); KOSPI 54% — FSS	6

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares *4 Money Today (4 Mar 2026, citing an FSS release): 2025 IPO lock-up commitments and pricing within the range *5 KSJ Research
Note: All figures in the answers come from the same calculations as the body

Assumptions and sources: data, sourced, judgment and design kept apart

Target Co., deal structure and Monte Carlo inputs — value, peer range, basis, status

30 inputs: data 9 · sourced 7 · verify 2 · design 3 · judgment 3 · needed 2 [*1,*2,*3]

Item	Value (range)	Unit	Basis / source	Status
Revenue FY2025A	3,000	KRW 100mn	Hypothetical company size, FY2025A — above the mid-size threshold; sets the price unit only	Design
Mid-size revenue threshold (ref.)	1,500	KRW 100mn	Framework Act on SMEs, Enforcement Decree Annex 1 (electrical equipment: KRW 150bn average revenue)	Verify
Revenue growth (FY2026E)	18.6% (13.7–24.7)	%	DART, 3-name combined revenue CAGR FY2022A→FY2025A (range = min–max)	Data
Operating margin	14.4% (8.6–24.4)	%	DART, 3-name combined operating margin FY2025A (range = min–max)	Data
Net margin (to parent)	10.3% (5.8–18.0)	%	DART, 3-name combined net margin to parent FY2025A (range = min–max)	Data
Net debt ÷ revenue	2.7% (-18.7–11.7)	%	DART, net debt ÷ revenue FY2025A (names with debt data; range = min–max)	Data
D&A ÷ revenue	—	%	Not collected → EV/EBIT as proxy	Input needed
Pre-IPO shares	10,000,000	shares	Hypothetical pre-IPO share count — changes the price level only	Design
Offer size (% of post-IPO shares)	25.0%	%	Offer as % of post-IPO shares — KRX listing rules' free-float requirement (e.g. 25% public holders)	Verify
Primary share (rest secondary)	80.0%	%	Primary share of the offer (rest secondary sale) — hypothetical deal design	Design
Lock-up (major holders)	—	months	Not set → overhang shown in shares only	Input needed
Band discount (low)	21.9% (14.5–28.8)	%	Public source	Sourced
Band discount (high)	35.0% (27.9–42.1)	%	Public source	Sourced
Growth mean (log)	17.5% (7.3–25.0)	log	DART, 9 annual revenue growth observations FY2023A–FY2025A (log mean)	Data
Growth s.d. (log)	6.28%	log	Same 9 observations (log standard deviation)	Data
1-yr-ahead P/E distribution	Triangular	-	Triangular: min conservative, mode normalised, max current — neutral shape without probability evidence	Neutral
Band low quantile	25.0%	%	Band low = 25th percentile of 1-yr value (25% loss probability at the low end)	Judgment
Band high quantile	50.0%	%	Band top = median 1-yr value (50% loss probability at the top)	Judgment
1-yr net margin min	3.1%	%	DART, 3-name combined net margin FY2022A (trough end)	Data
1-yr net margin mode	7.0%	%	DART, 3-name combined net margin over FY2022A–FY2025A (cycle average)	Data
1-yr net margin max	10.3% (5.8–18.0)	%	DART, 3-name combined net margin FY2025A (today = peak end)	Data
1-yr net margin distribution	Triangular	-	Triangular with the mode at the 4-yr cycle average = mean reversion; capped at today, floored at FY2022A	Judgment
Corr.: growth–P/E	0	-	Growth–P/E correlation 0: no data to estimate (neutral); co-movement tested as a sensitivity	Neutral
Corr.: growth–margin	0	-	Growth–margin correlation 0: no data to estimate (neutral)	Neutral
Corr.: margin–P/E	0	-	Margin–P/E correlation 0: neutral; in reality both may fall together, so the downside may be understated	Neutral
Institutional lock-up commitments 2025	41.0%	%	FSS release: institutional lock-up commitments 41% in 2025 (Money Today)	Sourced
Institutional lock-up commitments 2024	18.0%	%	FSS release: institutional lock-up commitments 18% in 2024 (Money Today)	Sourced
Institutional lock-up commitments, KOSPI 2025	54.0%	%	FSS release: KOSPI institutional lock-up commitments 54% in 2025 (Money Today)	Sourced
2025 IPOs priced within the range	100.0%	%	FSS release: every 2025 IPO priced within its range (Money Today)	Sourced
2025 H2 IPOs priced at the top	97.0%	%	FSS release: 97% of H2 2025 listings priced at the top of the range (Money Today)	Sourced

Source: *1 FSS DART annual reports (consolidated) *2 Framework Act on SMEs Enforcement Decree Annex 1; KRX listing rules (free-float requirement) *3 KSJ Research
Note: Status (same as Vol.2): data/sourced (gold) = DART or public source · judgment (shaded, brown) = unsourced setting, stress-tested · needed (red) = not set, fallback rule · verify = public basis, not yet checked · design = hypothetical setting · neutral = neutral shape without evidence

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Data notes: sources, dates and the SEC entity check

Series as-of 2026-10-20 · KRX 2026-10-19 · SEC 2026-10-20 · USD/KRW 1,332.1 (2026-10-20)

Data sources · datasets · dates

Source	Dataset / items	As of	Status
DART · FSS	Annual reports, consolidated, FY2022A–FY2025A (4 Korean names)	2026-10-20	Collected
KRX · Korea Exchange	Close, market cap, listed shares (2 Jan 2023 · 19 Oct 2026)	2026-10-20	Collected
SEC EDGAR	10-K annual financials (XBRL), cover shares, entity check	2026-10-20	Collected
yfinance (secondary)	U.S. peers' closes (first 2023 trading day · as-of)	2026-10-20	Collected
DART·SEC (extra)	D&A (for EBITDA) · SEC FY2022A NI	2026-10-20	Collected
ECOS · Bank of Korea	USD/KRW (display conversion)	2026-10-20	Collected
DART prospectuses	New-listing prospectuses: value, range, discount (n=148, filed 2024-12-03 to 2026-10-20)	2026-10-20	Collected

Note: Priority: official filings (DART, SEC) > public agencies (KRX, ECOS) > secondary libraries (yfinance, cross-checked with SEC shares) Check: DART corp_cls is today's listing status, so filtering on it drops firms that listed later (look-ahead) → classified at filing date (listing date > filing date, IPO wording); 119 filings restored.

SEC entity check, 5 names — 0 WARN; Vistra OK after checking the 10-K cover registrant [*1]

Company	Ticker	Listed CIK	Filer name (companyfacts)	Same group	Verdict
Vertiv	VRT	0001674101	Vertiv Holdings Co	yes	OK
Eaton	ETN	0001551182	EATON CORPORATION plc	yes	OK
GE Vernova	GEV	0001996810	GE Vernova Inc.	yes	OK
Constellation Energy	CEG	0001868275	CONSTELLATION ENERGY CORPORATION	yes	OK
Vistra	VST	0001692819	Vistra Operations Company LLC	yes (manual)	WARN → OK

Source: *1 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Market cap follows the listed entity (ticker map), financials the companyfacts filer — a mismatch drops the multiple (R3). Manual checks are recorded with URL and date; the automatic verdict is kept. Vistra source: https://www.sec.gov/Archives/edgar/data/1692819/000169281922000005/vistra-20211231.htm (checked 2026-10-22)

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Methodology: one rule set for every peer and date; conclusions from the numbers

Every figure is calculated from data and assumptions, and conclusions are generated from the results

Data → multiples → price → checks and report [*1]

1. Data

DART · KRX

SEC EDGAR

yfinance (sec.)

Entity, share checks

2. Multiples

3-step screen

Agg., mean, median

Outliers R1–R5

3 scenarios

3. Price

Equity value

Post-IPO shares

Discount grid

Sensitivity

4. Checks, report

MC seed, convergence

Multiple, unit, FX checks

Conclusion titles

Korean & English

Built with an AI coding tool (Claude Code) for data links, multiples, Monte Carlo and report generation; question design, peer and assumption judgments and number checks by the author.

Source: *1 KSJ Research
Note: Point-in-time rule: samples use only what was knowable then — an IPO is judged at the filing date (listing date > filing date, range and book-building wording), not by today's DART corp_cls. Full code and data sources: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Glossary: the language of IPO pricing

Units: KRW 100mn and KRW, USD mn · %, x

Glossary

Term	Definition
PER	Price-to-earnings = market cap ÷ NI to parent; the main IPO metric
Aggregate P/E	Σcap ÷ ΣNI — an earnings-weighted average that damps high multiples on small profits
EV/EBITDA	Enterprise value (cap + net debt) ÷ EBITDA; a capital-structure-neutral cross-check
EV/EBIT	Proxy without D&A: EV ÷ operating income (higher than EV/EBITDA)
Value per share	Equity value ÷ post-IPO shares — the price before discount
IPO discount	1 − offer price ÷ value per share; compensation for post-listing uncertainty
Indicative price range	The price range offered before book-building (low–top)
Primary · secondary	Primary = new shares (cash to company); secondary = existing holders sell (cash to them)
Overhang	Existing holders' shares that may hit the market after lock-ups
Lock-up	A period after listing during which major holders cannot sell
Three-step peer screen	Industry → financial tests (profit, multiple, filer) → business fit
Cycle peak	Earnings and multiples both high; risk that slowing profits and de-rating coincide
Monte Carlo	Repeated draws from input distributions to get an output distribution (fixed seed = reproducible)
P10 · P50 · P90	10th, 50th and 90th percentiles of the outcome
Loss probability	P(1-yr value per share < offer price)
Double peak	Margins and multiples both at a cycle peak — peak earnings × peak multiple
Lock-up commitment	An institution's pledge not to sell allotted shares for a set period

Note: Shaded = core IPO-pricing terms

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Disclosures

- This is a personal portfolio for education and research. It is not investment advice or a recommendation on any security.
- Target Co. is hypothetical and unrelated to any real company; the price range illustrates a method.
- Views are the author's own and do not represent any current or future employer.
- All data are public; accuracy and completeness are not guaranteed. Assumptions and sources are in Appendix A2.
- As of 2026-10-20. Methods and code can be reproduced from the repository.
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KSJ Research

https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

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